Chapter 4
Skyways in Minneapolis

4.1 Introduction

Often enclosed and climate controlled, skyways link second level corridors across buildings, connecting various activity hubs such as shops and offices (Robertson, 1994; Byers, 1998). Such links allow for more efficient movement of pedestrians while protecting them from weather and the hazards of vehicular traffic below. Among North American cities that have substantial skyway systems, Minneapolis witnessed the city's first skyway link in 1962 as a modest effort to provide greater access to the central business district (CBD) (Byers, 1998). Over the next four decades, a system of skyway links has emerged, resulting in a network that connected over 70 continuous blocks in downtown Minneapolis. Corbett et al. (2009) comprehensively recapitulates the evolutionary history of the Minneapolis skyway system.

Minneapolis's first skyway bridge connected the Northstar Center, downtown's first mixed-use building, with the Northwestern National Bank across Marquette Avenue in 1962. The next year, a connection to the Roanoke Building over 7th Street was also completed. The original intent of these connections, made possible by collaboration between the Minneapolis City Planning department and local business leaders, was to make the financial district "more convenient for business people and clients to traverse" (Jacob, 1984).

In 1969, three skyways were added to the system. One skyway linking the Radisson Hotel to the Radisson Mart and parking was built over 7th Street. Another skyway spanned 6th Street between the FirstStar Bank (Northstar Center) and the Rand Tower, connecting four blocks of the financial district. The 5th skyway connected Dayton's Department Store with the LaSalle court over 8th Street. This was the first skyway to connect establishments in the retail core (Kaufman, 1985; Byers, 1998).

Perhaps the most significant early addition to the system occurred when the IDS Center opened its doors in 1973. The block-sized, mixed-use complex featured four skyways that connected adjacent blocks in each direction. More importantly, the skyway over Nicollet Mall connected the financial district with the retail core, thereby establishing itself as the center of the system. It quickly became a new

F. Xie, D. M. Levinson, *Evolving Transportation Networks,*
Transportation Research, Economics and Policy,
DOI 10.1007/978-1-4419-9804-0_4, © Springer Science+Business Media, LLC 2011

landmark for the city and captured the imagination of city dwellers and additional business leaders regarding what skyways could do for downtown (Kaufman, 1985; Byers, 1998).

The construction of the early skyways drew much traffic and was deemed a tremendous success. Pedestrian traffic multiplied with the addition of more skyways, as did the number of merchants demanding space along these corridors. The property values of the second level rose, while values in the first-floor held up (Kaufman, 1985). As a result, the overall value of each connected building increased. By the early 1970s, retail space on the second level rented for as much as street-level space. Skyway connections were seen as an amenity to a building, thus owners could command higher lease rates for office space. As retail leasing rates steadily climbed throughout the 1980s, and in some cases were twice the value of some street level rates (Kaufman, 1985), owners and developers, fueled by the desire to add value to their properties, had new buildings designed and old ones retrofitted to include skyways.

By 1975, thirteen blocks within the retail and financial core of downtown were connected to the skyway system. From 1975 to 1985 the system grew at a rapid pace. Much of the development occurred at several sites on the fringes of the CBD (Byers, 1998). On the eastern edge of downtown, the Hennepin County Medical Center (HCMC) connected their buildings together, but was too far east to connect to the rest of the system. Substantial growth occurred in the Gateway District, just north of the CBD, linking office buildings and apartment high rises together. Additional skyways linked the Government Center to areas south and east including a municipal parking ramp, the Lutheran Brotherhood Building, and Centre Village. The new Piper Jaffray Building linked itself and the Energy Center to the main system near the financial core, and City Center, a shopping and entertainment complex completed in 1983, connected two additional blocks to the retail core.

During 1986-1995, the skyway system expanded along the South Mall to Orchestra Hall, and then onto the Convention Center. Significant skyway expansions were built to connect the Third Avenue parking ramps on the western edge of downtown with the retail and financial core of the system to entice auto travelers to patronize them (Byers, 1998). There were other skyways built, two of which provided better connectivity between the retail and financial cores, but most of the construction was sponsored by the City of Minneapolis government and accounted for fifteen blocks that were added to the system.

Between 1996 and 2004, the pace of skyway construction had slowed. A few blocks in the southwestern portion of the CBD were connected due to the new Target Headquarters and adjacent store and St. Thomas University expansion efforts. The completion of the Block-E retail/ entertainment center also occurred. Figures 4.1, 4.2, 4.3, 4.4, displaying the "snapshots" of the skyway system in 1975, 1985, 1995, and 2004, respectively, illustrate how the network has expanded in downtown Minneapolis over the past four decades.

As the system evolved, so did its governance. Unlike publicly owned skyway systems, such as those in Des Moines, Calgary, Cincinnati, and St. Paul, with their financing, construction, and management under direct city control (Montgomery

and Bean, 1999), the Minneapolis skyway system is a privately supplied network through which private buildings are connected to one another. In the early system, all of the links were privately built and operated, thus operating hours depended upon decisions made by the building owners on each side of a skyway connection. This created problems, especially in the evening hours when some workers and shoppers were not able to return to their cars the same way they came in. There were also concerns about skyway bridge standards and security. Since the 1980s, the city of Minneapolis started to regulate much of the skyway expansion. In 1980, the Minneapolis Downtown Council created the Skyway Advisory Committee[1] to set guidelines for minimum bridge widths, heights, and spans, and encourage owners to adopt uniform operating hours. The SAC provides design reviews and approvals for changes and additions to the skyway system, and serves in an advisory capacity to the Minneapolis City Council. Another important task includes setting standards for skyway system signage and navigational aids (Jacob, 1984).

Despite the increasing role played by governmental interests throughout the growth and development of the system, the evolution of the skyway network did not always align with what the City of Minneapolis had laid out in several visions of a skyway system connecting most of the blocks downtown (Jacob, 1984; Kaufman, 1985), leading some to criticize the seemingly haphazard growth (Byers, 1998). This chapter attempts to determine if the growth of the system followed a predictable path. One might posit that the system expanded to where it was valued the most. This may be reflected in the assessed worth of the surrounding unconnected blocks or perhaps in the number of people (i.e. jobs). In this research the point accessibility of each block is calculated to assess the values of a place before and after skyway construction; a connect-choice analysis is then carried out to determine if the accessibility measure can be used to predict the growth of the skyway system in downtown Minneapolis.

4.2 Methodology

4.2.1 Accessibility analysis

Accessibility measures the relative ease of reaching valued destinations (Hanson, 1995). To determine if accessibility could be used as a predictor of network growth, the point accessibility of each block lying within and adjacent to the connected skyway system had to be calculated right before each time the system was expanded. In this research, point accessibility for each of the blocks in downtown Minneapolis is determined using the following mathematical relation:

[1] The Committee consisted of 17 members that owned or occupied properties connected by skyways, plus six non-voting members including representatives from the City Coordinator's Office, Department of Public Works, City Planning Department, City Attorney's Office, Minneapolis Community Development Agency (MCDA) and the Department of Inspections.

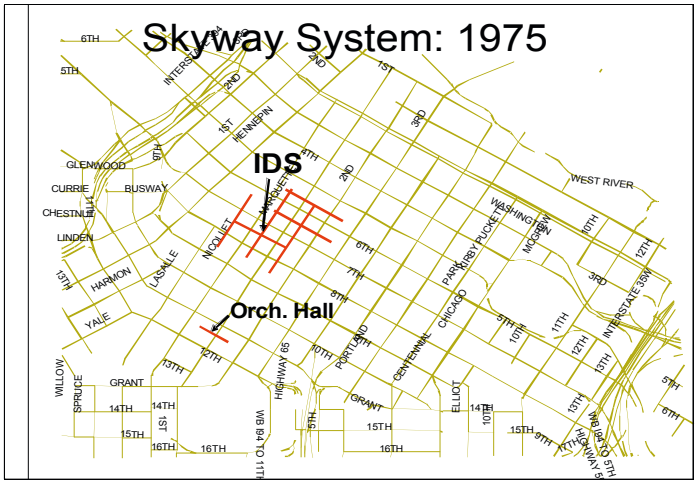


Fig. 4.1 The skyway network in downtown Minneapolis in 1975

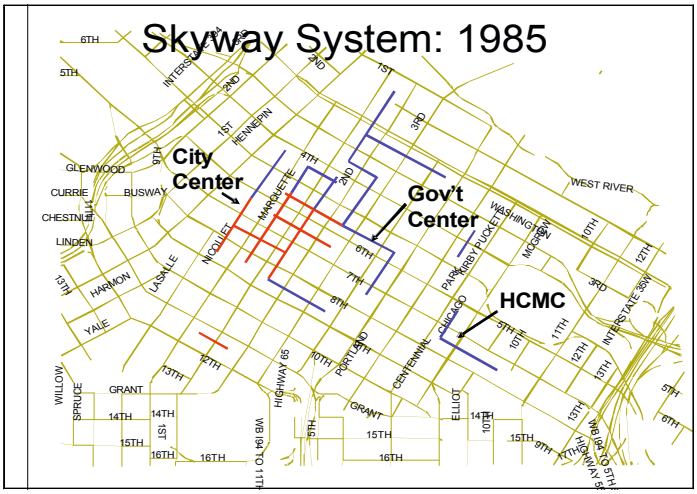


Fig. 4.2 The skyway network in downtown Minneapolis in 1985

$$A_i = P_i \sum_j P_j f(t_{ij}) \tag{4.1}$$

Where P_i and P_j represent some measure of activity, in this case, number of jobs held or number of trips generated in block i and block j, respectively; t_{ij} denotes the cost to travel between i and j along the shortest path, in this case, the least travel time walking from block i to block j; the cost function $f(t_{ij})$ is determined using a

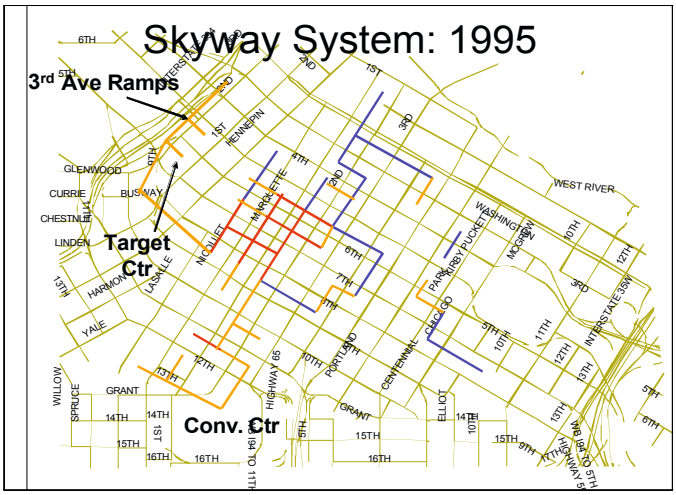


Fig. 4.3 The skyway network in downtown Minneapolis in 1995

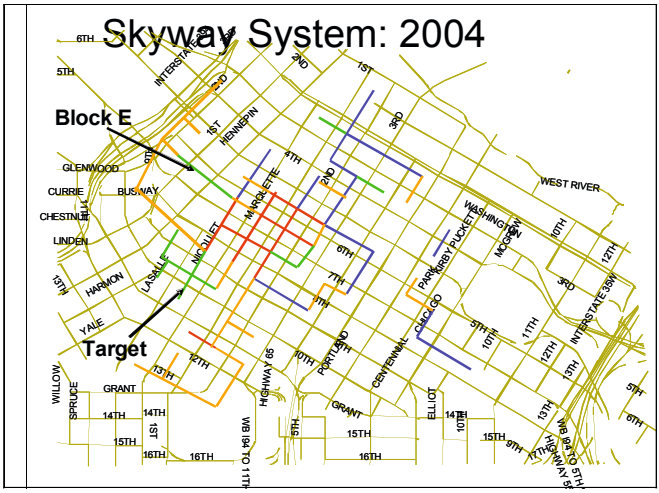


Fig. 4.4 The skyway network in downtown Minneapolis in 2004

gravity model. For simplicity, this analysis specifies a cost function in the following quadratic form:

$$f(t_{ij}) = \frac{1}{t_{ij}^2} \tag{4.2}$$

Measurement of point accessibility required estimating the number of job opportunities or number of trips at each block and the travel times between each pair of

the locations in the skyway system. The number of employees on each block was determined from the square footage of the buildings located there. Current data on square footage was obtained from Kramer (2004) and the City of Minneapolis and Hennepin County's Property Finder/Information websites. Historic maps and other books dealing with the history of downtown Minneapolis were used to determine what buildings were in place for each of the years the skyway system experienced growth.[2]

Trip generation rates were obtained from the Institute of Transportation Engineers (1997). Since most of the land use in downtown Minneapolis was devoted to office and retail, determining trip generation rates for each block was not difficult. The ITE Trip Generation Handbook did not have parking ramps listed as a land use, so parking ramp occupancy data was obtained from the City of Minneapolis (which owns most of the ramps) to estimate trip generation rates. The website for the City of Minneapolis stated there were currently 161,000 people working in downtown's 3.6 million m^2 (39 million ft^2) of office and government space. Using these numbers, an average of 22.5 m^2 (242 ft^2) per person was calculated and subsequently used to determine the number of employees on each block connected by the skyway system. Whether or not this average held true back to 1962 is debatable, but due to a lack of data in this area it seemed to be the best option.

It needs to be noted that, although point accessibility has been calculated separately using two measures of land use activity, namely number of jobs and number of trips, and used as the input to the connect-choice model explained below, the model was found to perform better with point accessibility calculated based on trip rates. It was not surprising as the activities that took place on parking ramps would otherwise be ignored if only number of jobs in each block is considered. To save space, only results generated based on estimated trip rates are reported in the next sections.

Estimation of travel times between each pair of the blocks (all blocks are considered origins and destinations) required several steps. First, a street map of Downtown Minneapolis in GIS format was downloaded from the Metropolitan Council's DataFinder website. Using ArcMap, the skyways were then manually added over the street grid as shown in Figure 4.5. The network under examination consists of 118 street segments and 73 skyway connections. For simplicity, it was assumed that all of the skyway connections were straight paths from the mid-block of one building to another. It was also assumed that connections from one block to the next that did not have connecting skyway links (street only) were straight paths from the center of one block to the center of the other block. Similarly, links that had the

[2] Gathering data on historic structures proved difficult. The property information websites often did not yield data on structures that no longer exist, even if their addresses were provided. Old maps usually showed the footprints of buildings, not the number of square feet or stories. Old photographs/illustrations were sometimes useful for estimating the square footage of buildings, but it was often difficult to estimate how much of the block these structures covered. In the absence of good data, educated guesses were made. In the cases where data on particular blocks were not available, it was assumed the blocks contained a collection of 3-5 story buildings that covered approximately 75% of the block. This translates into 28,000 m^2 (300,000 ft^2) and is comparable to the current makeup of many of the blocks in Minneapolis's Warehouse District.

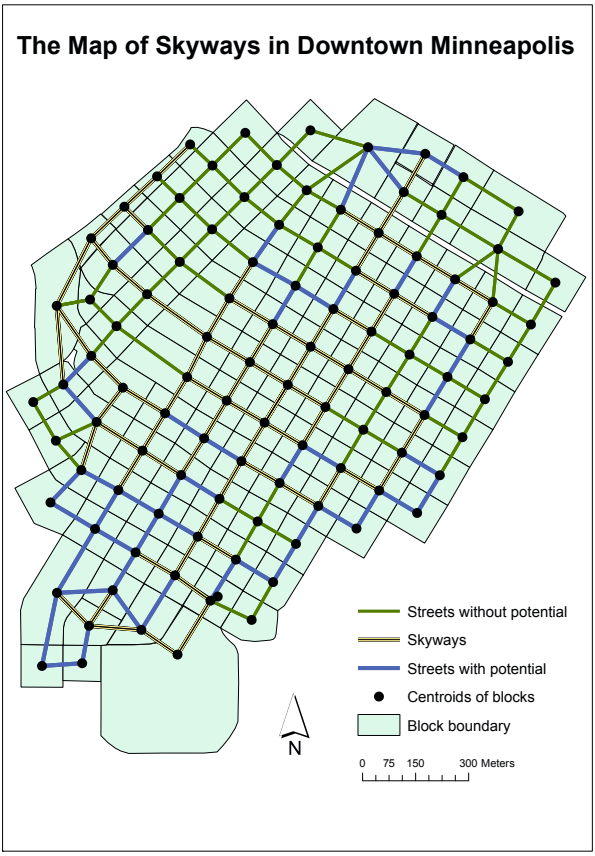


Fig. 4.5 The idealized skyway network in downtown Minneapolis

potential to be connected to the skyway system were added as links connecting the center of one block to the center of the next block. Next, nodes were created at the center of each linked block. The location of each node was determined from the ArcMap coordinates. The length of each link connecting one node to the next was then calculated according to node coordinates.[3]

Calculating the accessibility of each of the blocks involved an additional set of assumptions. The average walking speed was assumed to be 4.8 km/hr (3.0 mi/hr). However, in the case where travel had to be made from one block to another without using a skyway, the average walking speed was reduced to 2.4 km/hr (1.5 mi/hr).

[3] The map distance across a link does not necessarily represent the real distance one covers moving from one block to another. Between non-skyway blocks one has to go to the nearest intersection, and cross the street, thus increasing distance. The increased distance in the non-skyway case is accounted for by assuming a lower walking speed to cover the map distance of a link. See next paragraph for explanation.

This was done to account for the extra travel time and waiting time to cross streets (especially since most street crossing are made at the ends of blocks, rather than mid-block). Thus we essentially assumed the skyway link between two nodes has half the travel time cost of a non-skyway link. While to some extent these assumptions are arbitrary, they are unlikely to affect the accessibility rank of unconnected buildings, since all are treated equally.

Using these assumptions, point accessibilities were calculated for each block for each year the skyway system expanded. A connect-choice analysis was then conducted to determine if the most accessible blocks were in fact connected first.

4.2.2 Connect-choice analysis

In order to determine how often the expansion of the skyway network connected the blocks with higher accessibility, a connect-choice logit analysis was carried out relating the probability of an unconnected block's joining the network (in a given year) to its accessibility and current network size.

An algorithm was first developed to extract potential skyway connections in an iterative process, and evaluate the impact of building each candidate connection on accessibility. The flowchart of this algorithm is illustrated in Figure 4.6.

It is worth noting that the algorithm is generalizable to a variety of transportation networks. Its application to the Interurban network of Indiana will be discussed in the next chapter. In the case of the Minneapolis skyway system, the algorithm was run for 21 iterations, each of which represented a year when at least one skyway was built since 1962. Each iteration includes the following steps:

- Step 0: Find all candidate links for skyway construction given the network topology in the corresponding year of each iteration. All the links that have not been built as skyways by the year of examination and do not connect to any restricted blocks (i.e., historic districts and parks) are identified as candidates.
- Step 1: Change a candidate non-skyway link into a skyway link. The speed for non-skyway links, or street links is 2.4 km/hr (1.5 mi/hr), while the speed for skyway links is to 4.8 km/hr. After a candidate link is converted into a skyway link, the improved travel time between any pair of blocks in the network is recalculated using the shortest-path finding algorithm. The speed on this link is then restored to 2.4 km/hr (1.5 mi/hr) because the skyway construction is hypothetical.
- Step 2: Evaluate the accessibility impact. Based on the improved travel time upon hypothetically constructing a particular skyway link, the increase in accessibility for the two blocks that this link immediately connects to is calculated, as is the increase in accessibility for all the blocks in the network.
- Step 3: Implement the actual skyway connections. After evaluating all the candidate links one by one by repeating Step 1 and Step 2, the links where skyways were actually built in the corresponding year are updated with their improved travel speeds (and thus travel times) and labeled as "1" . The candidate links where no skyways were built in the corresponding year are labeled as "0".

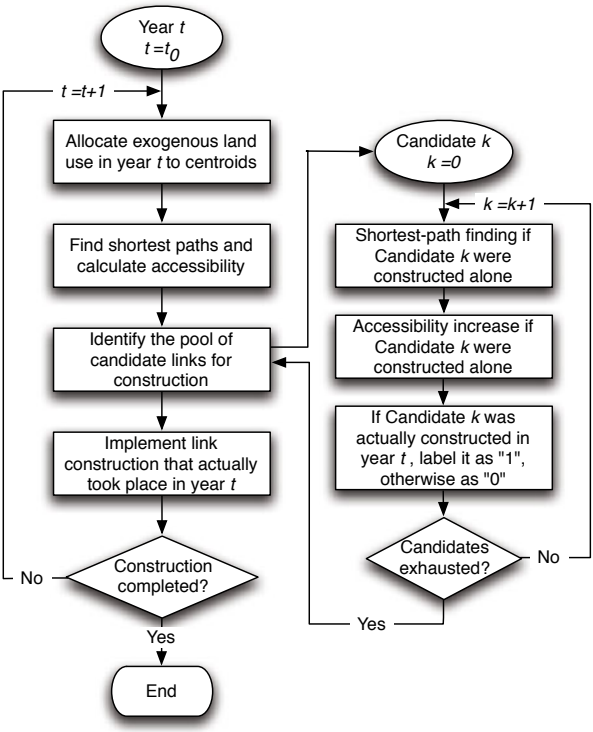


Fig. 4.6 Flowchart of the connect-choice analysis

Consequently, a dataset of 1,883 records was created, each representing a potential skyway connection for a particular year, with a binary variable ("0" or "1") indicating if this candidate connection was actually realized in this year. A logit model was then estimated using the statistical package STATA/IC 10.0 relating the binary variable to corresponding accessibility measures.

4.3 Results

As shown in Table 4.1, the connect-choice logit analysis indicates that the greater the increase in accessibility for the two connected blocks as a result of being connected to the network ($Ln(\Delta\tilde{A}_{i,j})$),[4] the more likely the candidate link will be built.

[4] Variable $\Delta A_{i,j}$ denotes the potential change in accessibility for Block i and Block j if the skyway connection between the two blocks were added to the network. Because of the large magnitude of this accessibility measure (out of 1,883 observations of $\Delta A_{i,j}$, 102 report a value of zero, indicating little effect of skyway construction on accessibility, while the remaining 1,781 measures range from

This is suggested by the estimated coefficient for $Ln(\Delta\tilde{A}_{i,j})$ that is both positive and statistically significant. The influence of $Ln(\Delta\tilde{A}_{-i,-j})$ (the increase in accessibility of the remaining blocks), on the other hand, is not significant, suggesting that skyway links were not built with the goal of improving the overall accessibility of the region in mind. This result could be explained by the fact that most of the skyway links are built between private buildings and paid for privately by the owners of these buildings, who constructed skyways primarily to serve their own interests, that is, to increase the commercial activities in their buildings and to improve the related land value, while not caring of others' interests. As $\Delta\tilde{A}_{i,j}$ and $\Delta\tilde{A}_{-i,-j}$ demonstrate strong correlation (the correlation coefficient is 0.72), the product of the two variables ($\Delta\tilde{A}_{i,j} \cdot \Delta\tilde{A}_{-i,-j}$) is also included in the model to control their cross-effects. Other than the accessibility measures, we have included three explanatory variables: *year* (indicating the year of examination) and *size* (denoting the total length of constructed skyways in the year of examination) to eliminate possible time trend effect of connect-choice, although we realized *year* and *size* may be highly correlated; and $size^2$ (total length squared) to test if the connect-choice has a non-linear (quadratic) relationship with the total length of the network. As shown in Table 4.1, none of the coefficient estimates before the three variables turned out to be statistically significant, suggesting the inclusion of them did not significantly influence the probablity of skyway construction. Above all, though the overall predictive power (as suggested by the pseudo R-squared of 0.18) is relatively low, the logit model reveals that the increase in local accessibility ($\Delta A_{i,j}$) is a significant predictor of network growth.

4.4 Findings and concluding remarks

This chapter has demonstrated in the case of the Minneapolis skyway system that the expansion of an urban transportation system has to some extent followed a predictable path. A connect-choice logit model relating the probability of joining the network (in a given year) to accessibility corroborates the hypothesis that accessibility is a significant explanatory factor of network growth, although the estimated statistical model is not determinative. The connect-choice analysis further discloses that the greater the increase in accessibility for the two connected blocks as a result of being connected by a candidate link, the more likely the link will be built. The overall accessibility improvement in the rest of the blocks, on the other hand, is not a significant predictor. This finding is consistent with the private supply of the Min-

6.21E3 to 1.01E8, averaging 5.29E6), this model includes the natural logarithm of the variable as an explanatory factor. Taking an logarithm directly on zeros, however, would eliminate the corresponding records from the data set. To address this issue, a common practice in statistics is to add a small positive value to ensure the positiveness of a variable such that its logarithm always exists. In this case, one unit is added to each accessibility measure and the updated variable is denoted as $\Delta\tilde{A}_{i,j}$ (i.e., $\Delta\tilde{A}_{i,j}=\Delta A_{i,j}+1$). Considering the large scale of the accessibility measure, this transformation will not affect model results significantly.

Table 4.1 Connect-choice logit model results for the Minneapolis skyway system

Number of observations = 1883
LR $\chi^2(5) = 95.91$
Prob $> \chi^2 = 0.00$
Log likelihood = -224.65
Pseudo $R^2 = 0.18$

variable	coefficient	std. error	z	P-Val.	95% Conf.	Interval
$Ln(\Delta\tilde{A}_{i,j})$	0.73	0.19	3.94	0.00	0.37	1.09
$Ln(\Delta\tilde{A}_{-i,-j})$	0.019	0.17	0.11	0.91	-0.32	0.35
$\Delta\tilde{A}_{i,j} \cdot \Delta\tilde{A}_{-i,-j}$	1.92E-15	1.03E-15	1.87	0.062	-9.43E-17	3.94E-15
size	-9.02E-08	2.18E-04	-0.00	1.00	-4.28E-04	4.28E-04
$size^2$	-6.15E-09	7.85E-09	-0.78	0.43	-2.15E-08	9.24E-09
year	0.057	0.060	0.94	0.35	-0.062	0.17
const	-126.64	118.60	-1.07	0.29	-359.09	105.81

neapolis skyway system, and suggests that network ownership plays a pivotal role in the development of an urban transportation system.

The network expanded to the blocks with high accessibility with only a few exceptions. One reason why the most accessible blocks were not always connected first may have to do with some of the specific physical characteristics of the buildings (i.e. connection difficulties and lack of logical entry points into the buildings). A number of building owners may also have been averse to the idea of losing rentable office space. More savvy building owners in less accessible locations may have felt the potential benefits of being connected were speculatively substantial and pushed aggressively to be included in the system. Many of the skyways connected to parking ramps, which generally had relatively low accessibility values due to the fact that most of them are located on the edges of downtown. In addition, when the Skyway Advisory Committee came into play as a regulator of skyway development since 1980, politics and redevelopment objectives also played a role, especially in the 1990s as the city sought to connect the Convention Center to the system (presumably to attract more conventions and bring more convention goers into the retail and restaurant areas) and the Third Avenue parking ramps (to encourage more patronage). This study could be extended to include more sophisticated considerations of ownership, alliance, and the conflicts between private and public interests during the development of the skyway network.